

INDUSTRIAL UNIVERSITY OF
HO CHI MINH CITY
FACULTY OF ELECTRONICS
TECHNOLOGY

FINAL EXAM

Course: Telecommunication System 2

Class:

Date:

Exam Duration: 90 minutes

(Excluding for distributing the exam papers)

Full name:; Student ID:

Question 1: (2 marks, CLO 1)

Describe the structure of the PSTN network and explain the functions of each component.

Why is a hierarchical architecture used in PSTN networks?

Question 2: (2 marks, CLO 1)

Present drawbacks and challenges of transmitting voice over IP networks.

Question 3: (2 marks, CLO 2)

Design an optimum configuration of a PABX for an enterprise. It is known that the enterprise have 30 department, each department has 4 offices. It is requirement that the QoS at 0.02. In the busy hour, the holding time is 3 minutes, the calling rate is 3 and about 10% out calling.

Question 4: (2 marks, CLO 3)

Cell planning for a mobile communication network of Ho Chi Minh city. It is known that the network uses GSM 900 standard, uplink 890-905MHz, downlink 935-950, the re-use pattern 4/12, QoS is 0.02. The area of Ho Chi Minh city is 2100km², population density is 1200 people per 1 km², in BH, the holding time is about 2 minutes, the calling rate is about 1. It is known that 10% out calling at BH.

Question 5: (2 marks, CLO 4)

Design a microwave propagation line between HCM and Vung Tau city. Assumed the distance between HCM and Vung Tau is 150km, the transceiver are parabolic antenna each 2m diameter, antenna' efficient is 0.55, the transmitting power is 20W, the carrier frequency is 2GHz, $\{S/N\}_{req}$ (dB) is 12dB, the received noise is $0.1\mu W$ and the earth radius to be 6400 Km.

----- End of Exam Paper -----

*Notice: - The exam allows the use of a hand-written A4 paper .
- The exam invigilator will not provide any further explanation.*